

JOB MARKET ANALYSIS

PRDA-04 | Data Analytics Project

Executive Summary

This project analyzes job-market trends for data-related roles using a dataset of 742 job postings from 343 companies. The analysis focuses on geographic demand, salary patterns, industries, companies, job titles, technical skills, education requirements, ownership type, and estimated salary ranges. Power BI was used to transform the prepared data into interactive dashboards and visual analysis.

The analysis shows that California has the highest number of job postings, while salary levels vary substantially by state, company, industry, role, and education. Data Scientist is the most frequently advertised role in the dataset. Excel, Python, and SQL appear among the important skills requested by employers, with cloud and visualization technologies also commonly represented.

The purpose of the project is not only to describe the job market, but also to translate the findings into practical recommendations that can help job seekers make informed decisions about location, role, skills, education, and target industries.

Project Objectives

- Identify states with the highest concentration of data-related job opportunities.
- Compare minimum, maximum, and average salary patterns across locations.
- Identify industries and companies with strong hiring demand and competitive salaries.
- Determine the most frequently advertised job titles and their salary levels.
- Examine technical skills requested by employers for different job titles.
- Explore the relationship between educational requirements and average salary.
- Assess job opportunities by company ownership type and estimated salary range.
- Present the findings through interactive Power BI dashboards and actionable recommendations.

Dataset Overview

Dataset Metric	Value
Job postings	742
Columns	42
Companies	343
Average salary	\$101.48K

The dataset contains job-posting attributes covering location, employer, industry, job title, salary estimates, skills, education, sector, ownership type, and related job characteristics. These fields support multi-dimensional analysis of demand and compensation across the data job market.

Data Cleaning and Preparation

Before visualization, the dataset required preparation to address missing and inconsistent values. The cleaning steps documented in the project presentation were:

- Removed a column containing only null values because it did not contribute useful analytical information.
- Replaced placeholder values of -1 with null values where appropriate and handled the affected records.
- Corrected values that fell outside the expected range.
- Reviewed the prepared fields before building the Power BI analysis.

Dashboard Overview

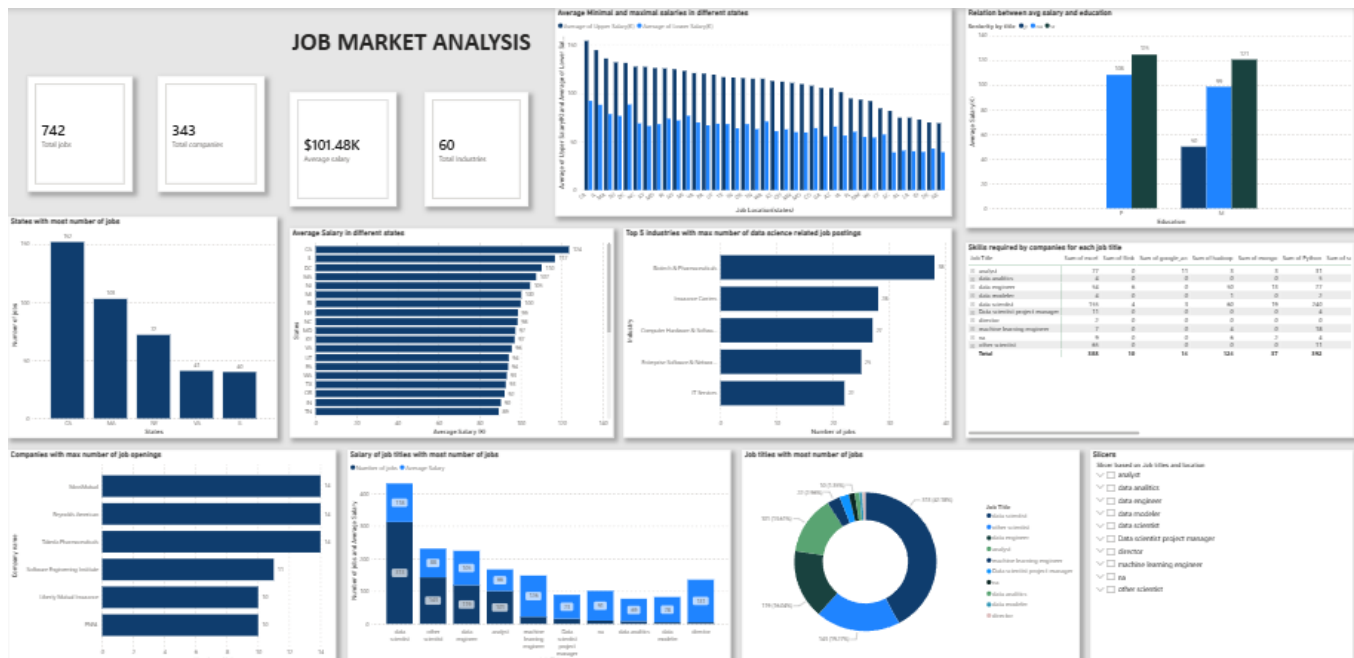


Figure 1. Main Job Market Analysis Dashboard

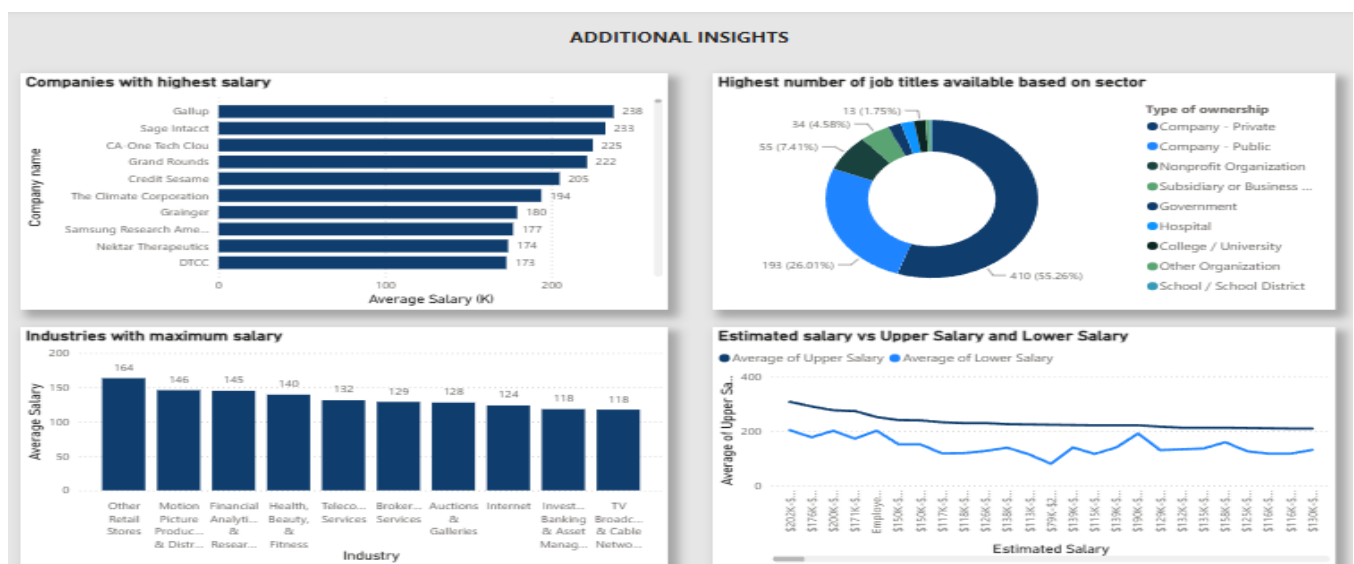
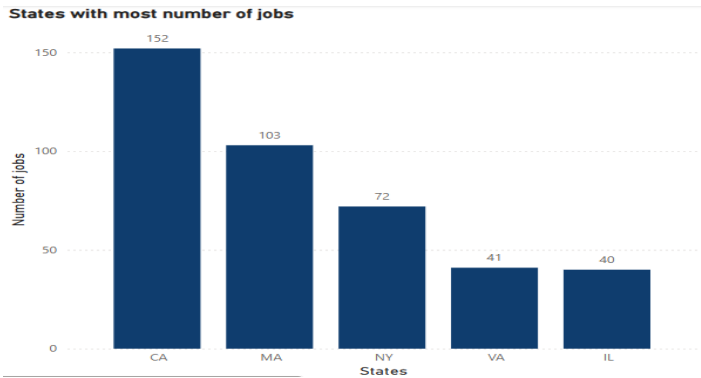


Figure 2. Additional Insights Dashboard

The dashboards consolidate the major job-market indicators and allow the analysis to be viewed across location, salary, company, industry, job title, skills, education, and other dimensions. Slicers were also used to support focused exploration by job title and location.

Analysis And Insights

States with the Most Number of Jobs

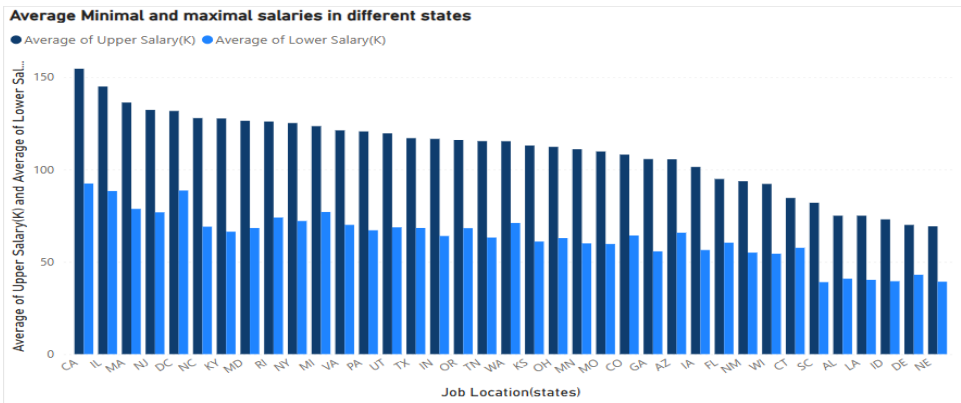


California (CA) has the highest number of job postings, followed by Massachusetts (MA) and New York (NY). This indicates that these states have a higher demand for data-related professionals and provide more job opportunities compared to other states

Recommendation:

Job seekers should identify the states with the highest number of job opportunities and focus their applications on those regions. As these states have higher demand, they should also be well prepared to face increased competition.

Average Minimum and Maximum Salaries in Different States

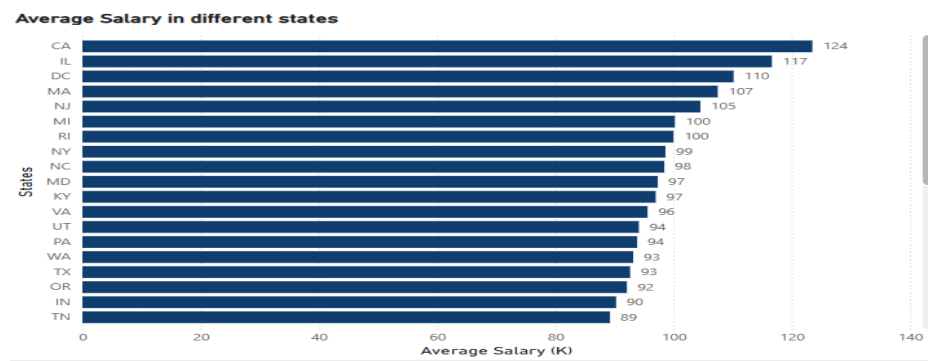


From the analysis, we can observe a significant difference between the average minimum and maximum salaries across different states. This suggests that salary is not determined only by the state but also depends on factors such as the job role, industry, company, and level of experience.

Recommendation:

Job seekers should compare the salary ranges offered for the same job role across different states before applying, helping them make better career decisions.

Average Salary in Different States

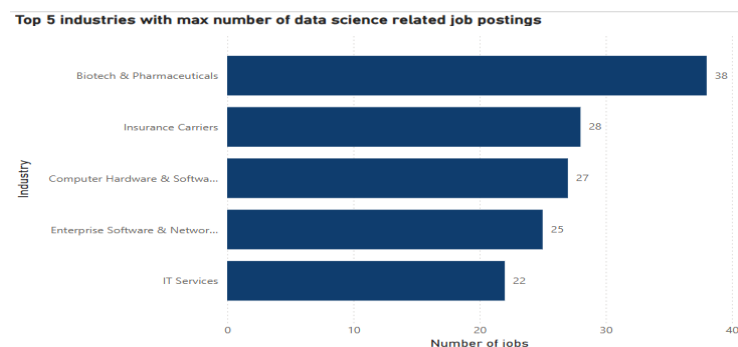


California (CA) has the highest average salary, followed by Illinois (IL) and the District of Columbia (DC). Although Massachusetts (MA) and New York (NY) have the highest number of job postings, their average salaries are lower than those of some other states. This shows that a higher number of job opportunities does not always mean higher average salaries.

Recommendation:

Job seekers should evaluate both salary levels and the availability of job opportunities when selecting locations, as states with more openings do not always offer the highest pay.

Top 5 Industries with the Maximum Number of Data Science-Related Job Postings

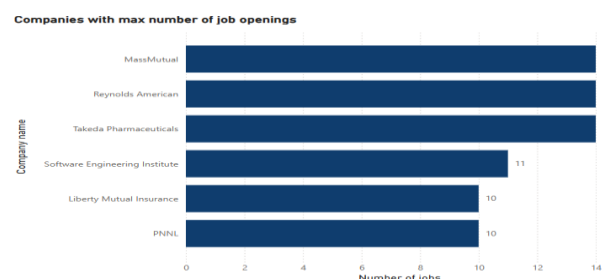


The Biotechnology & Pharmaceuticals industry has the highest number of data science-related job postings, followed by Insurance Carriers, Computer Hardware & Software, Enterprise Software, and IT Services. This shows that data science-related roles are no longer limited to IT companies and are becoming increasingly important across multiple industries.

Recommendation:

Job seekers should not limit their applications to only IT companies. They should explore opportunities across different industries because they also offer competitive data science roles.

Companies with the Maximum Number of Job Openings

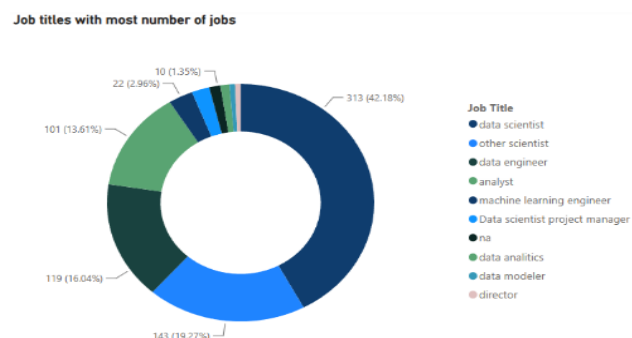


Companies such as MassMutual, Reynolds American, and Takeda Pharmaceuticals have the highest number of job openings. This indicates that data science-related roles are in high demand not only in IT companies but also in industries such as healthcare, insurance, and pharmaceuticals, reflecting the growth of data-driven decision-making.

Recommendation:

Job seekers should not limit their applications to IT companies. They should explore opportunities across industries as these sectors also have a strong demand for data science professionals which results to high job opportunity.

Job Titles with the Most Number of Jobs

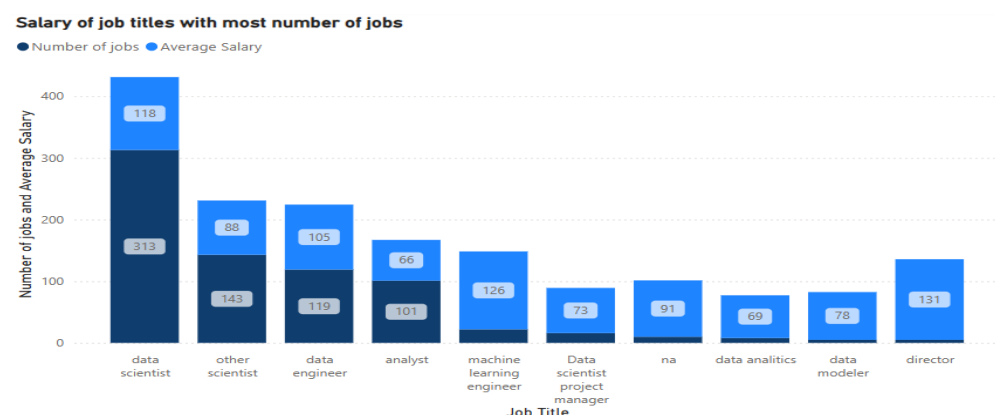


Data Scientist has the highest number of job postings, followed by Research Scientist and Data Engineer. This suggests that organizations are actively hiring professionals with strong analytical and technical skills

Recommendation:

As these roles have high demand and offer competitive salaries, developing the required technical skills can create better career opportunities in the data field.

Salary of Job Titles with Most Number of Jobs



Data Scientist has the highest average salary, followed by Research Scientist and Data Engineer. This is similar to the previous analysis, where these job titles also had the highest number of job postings. This shows that job roles with higher demand also tend to offer higher average salaries.

Recommendation:

Job seekers should stay updated with the latest job market trends and focus on high-demand roles, as these may provide more employment opportunities and offer competitive salaries.

Skills Required by Companies for Each Job Title

Skills required by companies for each job title							
Job Title	Sum of excel	Sum of flink	Sum of google_an	Sum of hadoop	Sum of mongo	Sum of Python	Sum of sc
analyst	77	0	11	3	3	31	
data analytics	4	0	0	0	0	5	
data engineer	54	6	0	50	13	77	
data modeler	4	0	0	1	0	2	
data scientist	155	4	3	60	19	240	
Data scientist project manager	11	0	0	0	0	4	
director	2	0	0	0	0	0	
machine learning engineer	7	0	0	4	0	18	
na	9	0	0	6	2	4	
other scientist	65	0	0	0	0	11	
Total	388	10	14	124	37	392	

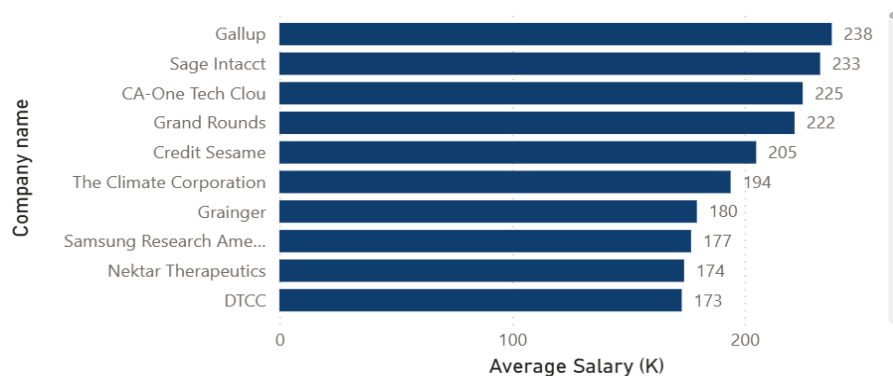
Excel, Python and SQL are among most important and required skills. Cloud technology and visualization tools are also commonly required indicating the importance of data science.

Recommendation

Job seekers should strengthen both their programming skills and data visualization skills to remain in competition.

Companies with the Highest Salary

Companies with highest salary

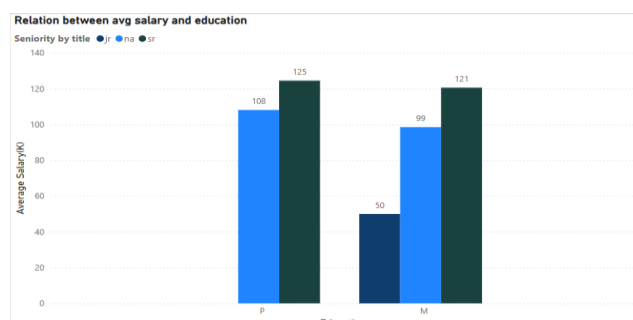


Gallup offers the highest average salary, followed by Sage Intacct and CA-One Tech Cloud. These companies provide some of the highest-paying opportunities for data professionals.

Recommendation:

Job seekers looking for higher salary packages should consider applying to these companies, but they should also be well prepared, as the competition for these roles is likely to be high.

Relation Between Average Salary and Education

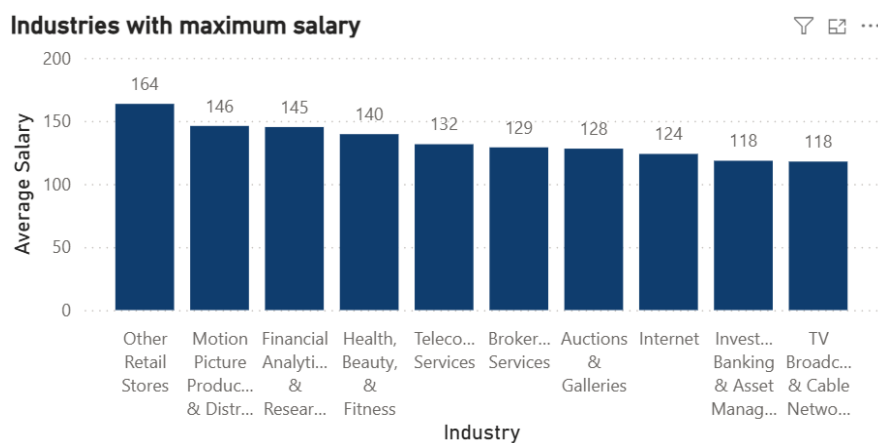


This chart indicates that P (PhD) has the highest average salary, followed by M (Master's degree). This shows that educational qualifications play an important role in salary, as jobs requiring higher levels of education generally offer better pay.

Recommendation:

Job seekers who aim for higher-paying roles should consider pursuing a Master's or PhD, as these qualifications are generally associated with better salary opportunities.

Industries with Maximum Salary

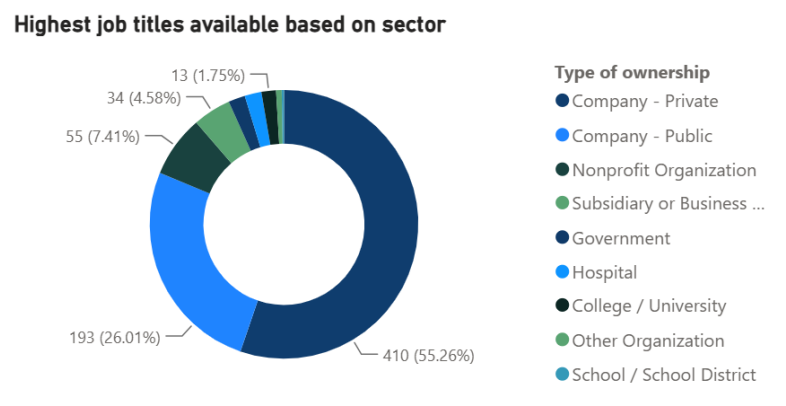


The Other Retail Stores industry offers the highest average salary, followed by Motion Picture Production & Distribution and Financial Analytics & Research. This indicates that high-paying data-related roles are available across different industries and are not limited to the technology sector. This also shows that job seekers can explore opportunities in different industries instead of focusing only on IT companies.

Recommendation:

Job seekers should not limit their applications to IT companies. They should explore opportunities across industries also as they have a strong demand for data science professionals which results to high job opportunity

Highest Job Opportunities Based on Type of Ownership



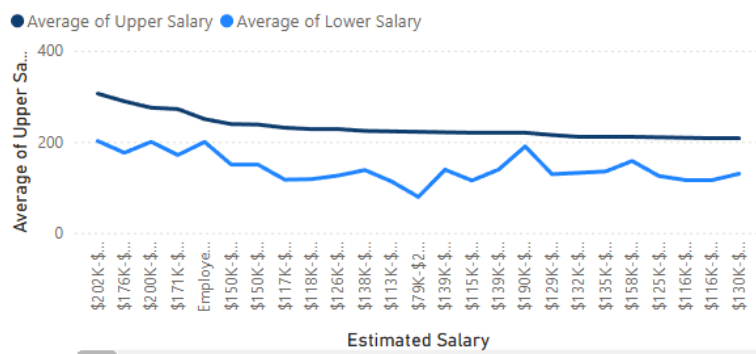
Private companies contribute the highest number of job opportunities, followed by public companies, while non-profit organizations and other ownership types contribute a much smaller share of job postings. This indicates that private and public companies are the major recruiters for data-related roles.

Recommendation:

Job seekers should also focus their job search on private and public companies, as they are the primary recruiters for data-related roles and provide greater employment opportunities with competitive salaries

Estimated Salary vs Upper Salary and Lower Salary

Estimated salary vs Upper Salary and Lower Salary



The chart shows that the average upper salary is always higher than the average lower salary across different estimated salary ranges. This reflects that salary varies based on factors such as the job role, company, skills, and experience. Candidates with better skills and qualifications are more likely to receive salaries closer to the upper range.

Recommendation:

Job seekers should compare both the lower and upper salary ranges before applying for a job instead of looking only at the estimated salary. They should also continue improving their skills to increase their chances of getting better salary packages.

Key Findings

- California records the highest job-posting volume, while Massachusetts and New York also show strong demand.
- High job volume and high average salary do not always occur in the same states.
- Data Scientist is the most frequently advertised job title in the dataset.
- Biotechnology & Pharmaceuticals leads the analyzed industries for data science-related postings.
- Excel, Python, and SQL are among the important employer-requested skills, with cloud and visualization tools also represented.
- Higher educational requirements are associated with higher average salaries in the analyzed data.
- Private companies account for the largest share of job opportunities by ownership type.
- High-paying data opportunities are distributed across several industries rather than being limited to technology.

Overall Recommendations

- Balance location decisions using both job availability and salary levels.

- Target high-demand roles such as Data Scientist, Research Scientist, and Data Engineer while building the skills associated with them.
- Develop a broad technical toolkit that combines analytical, programming, visualization, and cloud capabilities.
- Expand job searches beyond IT to sectors such as biotechnology, insurance, pharmaceuticals, retail, media, and financial analytics.
- Compare lower and upper salary ranges before applying rather than relying only on a single estimated salary value.
- Use education requirements strategically: advanced qualifications may support access to specialized higher-paying roles, but should be evaluated against career goals.

Conclusion

The Job Market Analysis shows that demand for data-related professionals varies significantly across states, industries, companies, and job titles. California leads the dataset in job volume, while compensation patterns differ by location and employer. Data Scientist is the most frequently advertised role, and employer requirements highlight the importance of Excel, Python, SQL, cloud technologies, and visualization skills.

Overall, the project demonstrates how Power BI can be used to convert job-posting data into clear, decision-oriented insights. By examining demand, salary, skills, education, company, industry, and ownership patterns together, the analysis provides a practical framework for job seekers to make more informed career and application decisions.